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METHOD FOR PLACING NON-REACTIVE COLLOID PARTICLES TO STOP GAS MIGRATION IN EXPANDABLE METAL APPLICATIONS

Abstract

Provided is a well system, and a method. The well system, in one aspect, includes a wellbore positioned within a subterranean formation, as well as a downhole tool positioned within the wellbore. The downhole tool, according to one aspect, includes a housing, as well as an expandable metal member positioned about the housing, the expandable metal member comprising a metal configured to expand in response to hydrolysis. The well system, in one further aspect, includes a reactive colloidal dispersion of colloid particles surrounding a surface of the expandable metal member.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 18/466,202, entitled “METHOD FOR PLACING NON-REACTIVE COLLOID PARTICLES TO STOP GAS MIGRATION IN EXPADABLE METAL APPLICATIONS”, filed on Sep. 13, 2023. The above-listed application is commonly assigned with the present application is incorporated herein by reference as if reproduced herein in its entirety.

BACKGROUND

[0002] Wellbores are drilled into the earth for a variety of purposes including accessing hydrocarbon bearing formations. A variety of downhole tools may be used within a wellbore in connection with accessing and extracting such hydrocarbons. Throughout the process, it may become necessary to isolate sections of the wellbore in order to create pressure zones. Downhole tools, such as frac plugs, bridge plugs, packers, and other suitable tools, may be used to isolate wellbore sections. Wellbore anchors may also be used to fix one or more downhole tools within the wellbore.

Description

BRIEF DESCRIPTION

[0003] Reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

[0004] FIG. 1 illustrates a perspective view of a well system including an exemplary operating environment that the apparatuses, systems and methods disclosed herein may be employed;

[0005] FIG. 2 illustrates one embodiment of a downhole tool designed, manufactured and/or operated according to one or more embodiments of the disclosure;

[0006] FIGS. 3A through 3C illustrate various different views of a downhole tool designed, manufactured and/or operated according to one or more embodiments of the disclosure coupled to a conveyance and deployed within a wellbore tubular;

[0007] FIG. 4 illustrates an alternative embodiment of a downhole tool designed, manufactured and/or operated according to one or more embodiments of the disclosure;

[0008] FIGS. 5A through 5C illustrate various different views of a downhole tool designed, manufactured and/or operated according to one or more embodiments of the disclosure coupled to a conveyance and deployed within a wellbore tubular; and

[0009] FIG. 6 illustrates a well system designed, manufactured and/or operated according to one or more alternative embodiments of the disclosure;

[0010] FIG. 7 illustrates one embodiment of a downhole tool designed, manufactured and/or operated according to one or more embodiments of the disclosure, as might be used in a well system such as the well system of FIG. 6;

[0011] FIGS. 8A through 8F illustrate various different views of a well system and downhole tool designed, manufactured and/or operated according to one or more embodiments of the disclosure at different stages of deployment; and

[0012] FIGS. 9A through 9J illustrate various different views of a well system and downhole tool designed, manufactured and/or operated according to one or more embodiments of the disclosure at different stages of deployment.

DETAILED DESCRIPTION

[0013] In the drawings and descriptions that follow, like parts are typically marked throughout the specification and drawings with the same reference numerals, respectively. The drawn figures are not necessarily to scale. Certain features of the disclosure may be shown exaggerated in scale or in somewhat schematic form and some details of certain elements may not be shown in the interest of clarity and conciseness. The present disclosure may be implemented in embodiments of different forms.

[0014] Specific embodiments are described in detail and are shown in the drawings, with the understanding that the present disclosure is to be considered an exemplification of the principles of the disclosure, and is not intended to limit the disclosure to that illustrated and described herein. It is to be fully recognized that the different teachings of the embodiments discussed herein may be employed separately or in any suitable combination to produce desired results.

[0015] Unless otherwise specified, use of the terms “connect,” “engage,” “couple,” “attach,” or any other like term describing an interaction between elements is not meant to limit the interaction to direct interaction between the elements and may also include indirect interaction between the elements described. Unless otherwise specified, use of the terms “up,” “upper,” “upward,” “uphole,” “upstream,” or other like terms shall be construed as generally away from the bottom, terminal end of a well; likewise, use of the terms “down,” “lower,” “downward,” “downhole,” or other like terms shall be construed as generally toward the bottom, terminal end of a well, regardless of the wellbore orientation. Use of any one or more of the foregoing terms shall not be construed as denoting positions along a perfectly vertical axis. Unless otherwise specified, use of the term “subterranean formation” shall be construed as encompassing both areas below exposed earth and areas below earth covered by water such as ocean or fresh water.

[0016] The present disclosure is based, at least in part, on the acknowledgment that the use of expandable metal in wellbore applications may be problematic under certain circumstances. For example, in certain instances once the expandable metal has encountered reactive fluid and expanded to expanded metal, fluid and/or gas may still migrate through the expanded metal. In one or more embodiments, the gas is able to migrate through interstitial spaces located in the expanded metal. Such issues may be problematic when the expandable metal is used in pressure isolation applications (e.g., frac plugs, bridge plugs, packers, plug and abandonments, etc.).

[0017] With this acknowledgment in mind, the present disclosure has recognized that by adding colloid particles to the resulting expanded metal, the gas migration problem discussed above may be significantly reduced and/or eliminated. In yet other embodiments, small colloid particles, extremely small colloid particles, excessively small colloid particles, or any combination thereof, may be added to the resulting expanded metal. The phrase “colloid particles,” as used herein, is intended to mean particles with a greatest dimension of no more than 500 nm. The phrase “small colloid particles,” as used herein, is intended to mean particles with a greatest dimension of no more than 300 nm. The phrase “extremely small colloid particles,” as used herein, is intended to mean particles with a greatest dimension of no more than 150 nm. The phrase “excessively small colloid particles,” as used herein, is intended to mean particles with a greatest dimension of no more than 50 nm. In at least one additional embodiment, all the colloid particles within the resulting expanded metal range from 1 nm to 500 nm, if not 1 nm to 100 nm, if not 5 nm to 50 nm.

[0018] The colloid particles, in at least one embodiment, are non-reactive colloid particles. For example, non-reactive clay particles could be used. In at least one other embodiment, the colloid particles are non-reactive polymer particles, such as latex particles. The latex can be a natural product or it can be formed through a synthetic reaction, such as an emulsion polymerization reaction. Latex is a colloidal dispersion and most latex consists of polymeric particles dispersed in a continuous medium. The latex can be synthesized from a range of monomers including acrylate (such as methyl methacrylate, butyl acrylate, ethylhexyl acrylate), styrene, vinyl acetate, and butadiene. Copolymers can also be used (such as styrene-acrylic, and styrene-butadiene).

Increasing the ratio of styrene ($T_g = +100^\circ \text{C.}$) to butadiene ($T_g = -85^\circ \text{C.}$) may be used to increase

the operating temperature of the latex. The ratio of styrene (high tensile strength) to butadiene (high elasticity) may also be used to influence the mechanical behavior of the resulting seal.

[0019] In yet another embodiment, the colloid particles are ionically charged colloid particles. The hydrolysis reaction that converts the expandable metal to expanded metal converts the expandable metal into a metal hydroxide. The multivalent metal ions of the expandable metal (Ca.sup.+2, Al.sup.+3, Mg.sup.+2) can induce the ionically charged colloid particles to become unstable and to allow the colloidal particles to pull together. Too much stability can prevent adhesion and filler acceptance, while too little stability can result in the colloid particles not remaining dispersed until use. The byproducts of the hydrolysis reaction will destabilize the colloid particles and will result in the colloid particles becoming a better filler precisely at the point that the filler properties are desired and at the point when the dispersion behaviors are no longer necessary. Stabilization can be added to the colloid particles in order to prolong the stability of the colloid particles. Stabilization commonly uses nonionic surfactants with hydrophilic chains (such as vinyl carboxylic acid, steric stabilizers, surfactants, etc.). Anionic and cationic surfactants can also be used.

[0020] The colloid particles may be included within the expanded metal in a number of different manners. For example, in at least one embodiment, the colloid particles may be positioned proximate the expandable metal in a powder form, but sometime prior to the expandable metal expanding to expanded metal, the powder of colloid particles encounter a fluid thereby forming a colloidal dispersion of colloid particles. This transition of the powder based colloid particles to the colloidal dispersion of colloid particles may occur downhole. For example, a dissolvable sleeve could hold the powder of colloid particles proximate the expandable metal until such time as the expandable metal is to undergo the hydrolysis, at which time it would mix with surrounding fluid to form the colloidal dispersion of colloid particles.

[0021] Those skilled in the art understand the different types of dissolvable materials that may be used for the dissolvable sleeve and remain within the scope of the disclosure. While a dissolvable sleeve has been discussed, other removable (e.g., slidable) sleeves may be used and remain within the scope of the disclosure. In at least one embodiment, the fluid is a reactive fluid, such that a reactive colloidal dispersion of colloid particles is achieved that would start the hydrolysis reaction.

[0022] In yet another embodiment, the colloid particles form a part of a non-reactive colloidal dispersion of colloid particles positioned proximate the expandable metal. Again, a sleeve (e.g., dissolvable sleeve) could hold the non-reactive colloidal dispersion of colloid particles proximate the expandable metal until such time as the expandable metal is to undergo the hydrolysis, at which time the colloid particles would mix with surrounding reactive fluid to form a reactive colloidal dispersion of colloid particles, thereby starting the hydrolysis reaction.

[0023] In yet another embodiment, a reactive colloidal dispersion of colloid particles may be pumped downhole until it encounters the expandable metal, at which time the hydrolysis reaction would begin. In such an embodiment, the reactive colloidal dispersion of colloid particles might fill an entirety of the wellbore above the expandable metal and at least a portion of the wellbore below the expandable metal. Nevertheless, in yet other embodiments, a small slug of the reactive colloidal dispersion of colloid particles could be pumped downhole (e.g., pinned between a downhole slug of unreactive fluid without the colloid particles and an uphole slug of fluid) until it encounters the expandable metal, at which time the hydrolysis reaction would begin.

[0024] In certain embodiments, the colloidal dispersion of colloid particles is a sturdy colloidal dispersion of colloid particles, or an extremely sturdy colloidal dispersion of colloid particles, or an excessively sturdy colloidal dispersion of colloid particles. The phrase "colloidal dispersion," as used herein, is intended to mean that less than 30 percent of the colloid particles would settle out of the dispersion within a 24 hour period. The phrase "sturdy colloidal dispersion," as used herein, is intended to mean that less than 15 percent of the colloid particles would settle out of the dispersion within a 24 hour period. The phrase "extremely sturdy colloidal dispersion," as used herein, is intended to mean that less than 10 percent of the colloid particles would settle out of the dispersion

within a 24 hour period. The phrase “excessively sturdy colloidal dispersion,” as used herein, is intended to mean that less than 5 percent of the colloid particles would settle out of the dispersion within a 24 hour period.

[0025] In certain embodiments, the mass of the colloidal particles can be less than or equal to the mass of the expandable metal in the pre-expanded form. In some embodiments, the mass of the colloidal particles ranges from 1% to 50% of the mass of the expandable metal in the pre-expanded form. The phrase “mass of the colloidal particles,” as used herein, is intended to mean the mass of the solids in the colloidal suspension, or alternatively the mass of the powder of colloidal particles. The phrase “a low colloidal mass,” as used herein, is intended to mean that the mass of the colloidal particles is less than 25% of the mass of the expandable metal in the pre-expanded form. The phrase “a very low colloidal mass,” as used herein, is intended to mean that the mass of the colloidal particles is less than 8% of the mass of the expandable metal in the pre-expanded form. The phrase “an extremely low colloidal mass,” as used herein, is intended to mean that the mass of the colloidal particles is less than 2% of the mass of the expandable metal in the pre-expanded form.

[0026] Referring to FIG. 1, depicted is a perspective view of a well system **100** including an exemplary operating environment that the apparatuses, systems and methods disclosed herein may be employed. For example, the well system **100** could use a downhole tool including expandable metal according to any of the embodiments, aspects, applications, variations, designs, etc. disclosed in the following paragraphs. The well system **100** illustrated in FIG. 1 includes a drilling rig **110** extending over and around a wellbore **120** formed in a subterranean formation **130**. As those skilled in the art appreciate, the wellbore **120** may be fully cased, partially cased, or an open hole wellbore. In the illustrated embodiment of FIG. 1, the wellbore **120** is partially cased, and thus includes a cased region **140** and an open hole region **145**. The cased region **140**, as is depicted, may employ casing **150** that is held into place by cement **160**.

[0027] The well system **100** illustrated in FIG. 1 additionally includes a downhole conveyance **170** deploying a downhole tool **180** within the wellbore **120**. The downhole conveyance **170** can be, for example, tubing-conveyed, wireline, slickline, work string, or any other suitable means for conveying the downhole tool **180** into the wellbore **120**. In one particular advantageous embodiment, the downhole conveyance **170** is American Petroleum Institute “API” pipe.

[0028] The downhole tool **180**, may comprise many different downhole tools employing expandable metal and remain within the scope of the disclosure. In the illustrated embodiment, however, the downhole tool **180** includes a housing, as well as an expandable metal member positioned about the housing, the expandable metal member comprising a metal configured to expand in response to hydrolysis. The downhole tool, in the illustrated embodiment of FIG. 1, may additionally include colloid particles surrounding a surface of the expandable metal member.

[0029] The term expandable metal, as used herein, refers to the expandable metal in a pre-expansion form. Similarly, the term expanded metal, as used herein, refers to the resulting expanded metal after the expandable metal has been subjected to reactive fluid, as discussed below. The expanded metal, in accordance with one or more aspects of the disclosure, comprises a metal that has expanded in response to hydrolysis. In certain embodiments, the expanded metal includes residual unreacted metal. For example, in certain embodiments the expanded metal is intentionally designed to include the residual unreacted metal. The residual unreacted metal has the benefit of allowing the expanded metal to self-heal if cracks or other anomalies subsequently arise, or for example to accommodate changes in the tubular or housing diameter due to variations in temperature and/or pressure. Nevertheless, other embodiments may exist wherein no residual unreacted metal exists in the expanded metal. In at least one embodiment, the residual unreacted metal exists when the expandable metal has expanded into contact with another feature, such as another wellbore tubular, prior to all of the expanded metal reacting into expanded metal. Once the expanded metal has sealed against this wellbore tubular, the reactive fluid may no longer reach the expandable metal, and the hydrolysis essentially ends. Similarly, if all of the expandable metal has

reacted into expanded metal prior to sealing against the wellbore tubular, the expanded metal may ultimately wash away.

[0030] The expandable metal, in some embodiments, may be described as expanding to a cement like material. In other words, the expandable metal goes from metal to micron-scale particles and then these particles expand and lock together to, in essence, seal two or more surfaces together. The reaction may, in certain embodiments, occur in less than 2 days in a reactive fluid and in certain temperatures. Nevertheless, the time of reaction may vary depending on the reactive fluid, the expandable metal used, the downhole temperature, surface-area-to-volume ratio (SA:V) of the expandable metal, etc.

[0031] In some embodiments, the reactive fluid may be a brine solution such as may be produced during well completion activities, and in other embodiments, the reactive fluid may be one of the additional solutions discussed herein (e.g., water-based mud). The expandable metal is electrically conductive in certain embodiments. The expandable metal, in certain embodiments, has a yield strength greater than about 8,000 psi, e.g., 8,000 psi $\pm$ 50%. The expandable metal, in at least one embodiment, has a minimum dimension greater than about 1.25 mm (e.g., approximately 0.05 inches).

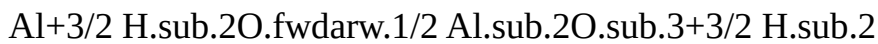
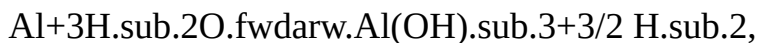
[0032] The hydrolysis of the expandable metal can create a metal hydroxide. The formative properties of alkaline earth metals (Mg—Magnesium, Ca—Calcium, etc.) and transition metals (Zn—Zinc, Al—Aluminum, etc.) under hydrolysis reactions demonstrate structural characteristics that are favorable for use with the present disclosure. Hydration results in an increase in size from the hydration reaction and results in a metal hydroxide that can precipitate from the fluid.

[0033] It should be noted that the starting expandable metal, unless otherwise indicated, is not a metal oxide (e.g., an insulator). In contrast, the starting expandable metal has, in certain embodiments, the properties of traditional metals: 1) Highly conductive to both electricity and heat (e.g., greater than 1,000,000 siemens per meter); 2) Contains a metallic bond (e.g., the outermost electron shell of each of the metal atoms overlaps with a large number of neighboring atoms). As a consequence, the valence electrons are allowed to move from one atom to another and are not associated with any specific pair of atoms. This gives metals their conductive nature; 3) Malleable and ductile, for example deforming under stress without cleaving; and 4) Tend to be shiny and lustrous with high density. In contrast, metal oxides are ceramics, and are dull, insulating, fragile, brittle and are not conductive metals.

[0034] The hydration reactions for magnesium is:



where $\text{Mg}(\text{OH})_2$ is also known as brucite. Another hydration reaction uses aluminum hydrolysis. The reaction forms a material known as Gibbsite, bayerite, boehmite, aluminum oxide, and norstrandite, depending on form. The possible hydration reactions for aluminum are:



Magnesium hydroxide is considered to be relatively insoluble in water. Aluminum hydroxide can be considered an amphoteric hydroxide, which has solubility in strong acids or in strong bases. Alkaline earth metals (e.g., Mg, Ca, etc.) work well for the expandable metal, but transition metals (Al, etc.) also work well for the expandable metal. In one embodiment, the metal hydroxide is dehydrated by the swell pressure to form a metal oxide.

[0035] In at least one embodiment, the expandable metal is a non-graphene based expandable metal. By non-graphene based material, it is meant that it does not contain graphene, graphite,

graphene oxide, graphite oxide, graphite intercalation, or in certain embodiments, compounds and their derivatized forms to include a function group, e.g., including carboxy, epoxy, ether, ketone, amine, hydroxy, alkoxy, alkyl, aryl, aralkyl, alkaryl, lactone, functionalized polymeric or oligomeric groups, or a combination comprising at least one of the forgoing functional groups. In at least one other embodiment, the expandable metal does not include a matrix material or an exfoliatable graphene-based material. By not being exfoliatable, it means that the expandable metal is not able to undergo an exfoliation process. Exfoliation as used herein refers to the creation of individual sheets, planes, layers, laminae, etc. (generally, “layers”) of a graphene-based material; the delamination of the layers; or the enlargement of a planar gap between adjacent ones of the layers, which in at least one embodiment the expandable metal is not capable of.

[0036] In yet another embodiment, the expandable metal does not include graphite intercalation compounds, wherein the graphite intercalation compounds include intercalating agents such as, for example, an acid, metal, binary alloy of an alkali metal with mercury or thallium, binary compound of an alkali metal with a Group V element (e.g., P, As, Sb, and Bi), metal chalcogenide (including metal oxides such as, for example, chromium trioxide, $\text{PbO} \cdot 2\text{H}_2\text{O}$, $\text{MnO} \cdot 2\text{H}_2\text{O}$, metal sulfides, and metal selenides), metal peroxide, metal hyperoxide, metal hydride, metal hydroxide, metals coordinated by nitrogenous compounds, aromatic hydrocarbons (benzene, toluene), aliphatic hydrocarbons (methane, ethane, ethylene, acetylene, n-hexane) and their oxygen derivatives, halogen, fluoride, metal halide, nitrogenous compound, inorganic compound (e.g., trithiazyl trichloride, thionyl chloride), organometallic compound, oxidizing compound (e.g., peroxide, permanganate ion, chlorite ion, chlorate ion, perchlorate ion, hypochlorite ion, As_2O_5 , N_2O_5 , $\text{CH}_3\text{DIO} \cdot 4\text{H}_2\text{O}$, $(\text{NH}_4)_2\text{S}_2\text{O}_8$, chromate ion, dichromate ion), solvent, or a combination comprising at least one of the foregoing. Thus, in at least one embodiment, the expandable metal is a structural solid expanded metal, which means that it is a metal that does not exfoliate and it does not intercalate. In yet another embodiment, the expandable metal does not swell by sorption.

[0037] In an embodiment, the expandable metal used can be a metal alloy. The expandable metal alloy can be an alloy of the base expandable metal with other elements in order to either adjust the strength of the expandable metal alloy, to adjust the reaction time of the expandable metal alloy, or to adjust the strength of the resulting metal hydroxide byproduct, among other adjustments. The expandable metal alloy can be alloyed with elements that enhance the strength of the metal such as, but not limited to, Al—Aluminum, Zn—Zinc, Mn—Manganese, Zr—Zirconium, Y—Yttrium, Nd—Neodymium, Gd—Gadolinium, Ag—Silver, Ca—Calcium, Sn—Tin, and Re—Rhenium, Cu—Copper. In some embodiments, the expandable metal alloy can be alloyed with a dopant that promotes corrosion, such as Ni—Nickel, Fe—Iron, Cu—Copper, Co—Cobalt, Ir—Iridium, Au—Gold, C—Carbon, Ga—Gallium, In—Indium, Mg—Mercury, Bi—Bismuth, Sn—Tin, and Pd—Palladium. The expandable metal alloy can be constructed in a solid solution process where the elements are combined with molten metal or metal alloy. Alternatively, the expandable metal alloy could be constructed with a powder metallurgy process. The expandable metal can be cast, forged, extruded, sintered, welded, mill machined, lathe machined, stamped, eroded or a combination thereof. The metal alloy can be a mixture of the metal and metal oxide. For example, a powder mixture of aluminum and aluminum oxide can be ball-milled together to increase the reaction rate.

[0038] Optionally, non-expanding components may be added to the starting metallic materials. For example, ceramic, elastomer, plastic, epoxy, glass, or non-reacting metal components can be embedded in the expandable metal or coated on the surface of the expandable metal. In yet other embodiments, the non-expanding components are metal fibers, a composite weave, a polymer ribbon, or ceramic granules, among others. In one variation, the expandable metal is formed in a serpentine reaction, a hydration and metamorphic reaction. In one variation, the resultant material resembles a mafic material. Additional ions can be added to the reaction, including silicate, sulfate, aluminate, carbonate, and phosphate. The metal can be alloyed to increase the reactivity or to

control the formation of oxides.

[0039] The expandable metal can be configured in many different fashions, as long as an adequate volume of material is available for supporting the necessary features. For example, the expandable metal may be formed into a single long member, multiple short members, rings, among others. In another embodiment, the expandable metal may be formed into a long wire of expandable metal, which can be in turn be wound around a housing as a sleeve. The wire diameters do not need to be of circular cross-section, but may be of any cross-section. For example, the cross-section of the wire could be oval, rectangle, star, hexagon, keystone, hollow braided, woven, twisted, among others, and remain within the scope of the disclosure. In certain other embodiments, the expandable metal is a collection of individual separate chunks of the metal held together with a binding agent. In yet other embodiments, the expandable metal is a collection of individual separate chunks of the metal that are not held together with a binding agent, but held in place using one or more different techniques, including an enclosure (e.g., an enclosure that could be crushed to expose the individual separate chunks to the reactive fluid), a cage, etc.

[0040] Additionally, a delay coating or protective layer may be applied to one or more portions of the expandable metal to delay the expanding reactions. In one embodiment, the material configured to delay the hydrolysis process is a fusible alloy. In another embodiment, the material configured to delay the hydrolysis process is a eutectic material. In yet another embodiment, the material configured to delay the hydrolysis process is a wax, oil, or other non-reactive material. The delay coating or protective layer may be applied to any of the different expandable metal configurations disclosed above.

[0041] Turning to FIG. 2, illustrated is one embodiment of a downhole tool **200** designed, manufactured and/or operated according to one or more embodiments of the disclosure. The downhole tool **200**, in the illustrated embodiment, has a length (l) and a width (w). The downhole tool **200**, in the illustrated embodiment, further includes a housing **210**. The housing **210**, in one or more embodiments, may be a tubular housing or a solid housing, regardless of cross-sectional shape. The housing **210** may additionally be a mandrel. Moreover, the housing **210** may comprise many different materials, including metal, and remain within the scope of the disclosure.

[0042] The downhole tool **200**, in the embodiment of FIG. 2, additionally includes an expandable metal member **220** positioned radially about the housing **210**, the expandable metal member **220** comprising a metal configured to expand in response to hydrolysis. In at least one embodiment, the expandable metal member **220** is placed radially outside of the housing **210**, such as is the case if the housing **210** is a mandrel. In yet another embodiment, the expandable metal member **220** may be placed radially inside of the housing **210**, such as is the case if the housing **210** is a tubular. The expandable metal member **220** may include any of the expandable metal member materials and configurations disclosed above. Nevertheless, in the embodiment of FIG. 2, the expandable metal member **220** is a single long member positioned between a pair of end rings **230** (e.g., metal end rings).

[0043] In accordance with one embodiment of the disclosure, colloid particles **240** surround a surface of the expandable metal member **220**. In at least one embodiment, the colloid particles **240** surround a radial outer surface of the expandable metal member **220**. However, in other embodiments, the colloid particles **240** surround a radial inner surface of the expandable member **220**. In yet other embodiments, the colloid particles surround another non-radial surface of the expandable member **220**. In the illustrated embodiment, the colloid particles **240** form a part of a non-reactive colloidal dispersion of colloid particles **250**, as discussed above. For example, the colloid particles **240** may be dispersed within a non-reactive fluid, such as oil or oil based mud. Furthermore, in the illustrated embodiment, a dissolvable sleeve **260** holds the non-reactive colloidal dispersion of colloid particles **250** proximate the expandable metal member **220** until such time as the expandable metal member **220** is to undergo the hydrolysis.

[0044] Turning now to FIGS. 3A through 3C, illustrated are various different views of a downhole

tool **300** designed, manufactured and/or operated according to one or more embodiments of the disclosure coupled to a conveyance **380** and deployed within a wellbore tubular **390**. The downhole tool **300** is similar in many respects to the downhole tool **200** of FIG. 2. Accordingly, like reference numbers have been used to indicate similar, if not identical, features.

[0045] The conveyance **380** may be any known or hereafter discovered conveyance and remain within the scope of the disclosure. The wellbore tubular **390** may also be any known or hereafter discovered wellbore tubular and remain within the scope of the disclosure. In at least one embodiment, however, the wellbore tubular **390** is wellbore casing or an uncased section of a wellbore.

[0046] In the embodiment of FIG. 3A, the dissolvable sleeve **260** remains intact, and thus keeps the colloidal particles (e.g., the non-reactive colloidal dispersion of colloid particles **250**) proximate the expandable metal member **220**. The dissolvable sleeve **260**, may additionally keep any potentially reactive fluids from reaching the expandable metal member **220**. However, in the embodiment of FIG. 3A, no reactive fluid is surrounding the downhole tool **300**.

[0047] Turning to FIG. 3B, illustrated is the downhole tool **300** of FIG. 3A after the downhole tool **300** has been subjected to fluid (e.g., reactive fluid **310** in one embodiment). The reactive fluid **310**, over time, may dissolve the dissolvable sleeve **260**. When the dissolvable sleeve **260** dissolves, colloid particles **240** from the non-reactive colloidal dispersion of colloid particles **250** may disperse within the reactive fluid **310** surrounding the downhole tool **300**, as shown.

[0048] Turning to FIG. 3C, illustrated is the downhole tool **300** of FIG. 3B after the reactive fluid **310** having the colloid particles **240** therein reacts with the expandable metal member **220**. The reactive fluid **310**, in the illustrated embodiment, causes the expandable metal member **220** to expand in response to hydrolysis, and thereby form an expanded metal member **330**. In accordance with the disclosure, given that the reactive fluid **310** now includes the colloid particles **240**, the expanded metal member **330** includes the colloid particles **240** in interstitial spaces thereof.

Accordingly, the expanded metal member **330** having the colloid particles **240** is more impervious to fluid and/or gas migration therethrough, as compared to a traditional expanded metal member.

[0049] Turning now to FIG. 4, illustrated is an alternative embodiment of a downhole tool **400** designed, manufactured and/or operated according to one or more embodiments of the disclosure. The downhole tool **400** is similar in many respects to the downhole tool **200** of FIG. 2.

Accordingly, like reference numbers have been used to indicate similar, if not identical, features. The downhole tool **400** differs, for the most part, from the downhole tool **200**, in that the downhole tool **400** employs colloid particles **440** in the form of a powder of colloidal particles **450**, as opposed to the non-reactive colloidal dispersion of colloid particles **250** discussed with regard to FIG. 2.

[0050] Turning now to FIGS. 5A through 5C, illustrated are various different views of a downhole tool **500** designed, manufactured and/or operated according to one or more embodiments of the disclosure coupled to a conveyance **580** and deployed within a wellbore tubular **590**. The downhole tool **500** is similar in many respects to the downhole tool **400** of FIG. 4. Accordingly, like reference numbers have been used to indicate similar, if not identical, features.

[0051] Turning to FIG. 5A, the dissolvable sleeve **260** remains intact, and thus keeps the powder of colloid particles **450** proximate the expandable metal member **220**. The dissolvable sleeve **260**, may additionally keep any potentially reactive fluids from reaching the expandable metal member **220**.

[0052] Turning to FIG. 5B, illustrated is the downhole tool **500** of FIG. 5A after the downhole tool **500** has been subjected to fluid (e.g., reactive fluid **510** in one embodiment). The reactive fluid **510**, over time, may dissolve the dissolvable sleeve **260**. When the dissolvable sleeve **260** dissolves, the colloid particles from the powder of colloid particles **450** may disperse within the reactive fluid **510** surrounding the downhole tool **500**, as shown.

[0053] Turning to FIG. 5C, illustrated is the downhole tool **500** of FIG. 5B after the reactive fluid

510 having the colloid particles **440** therein reacts with the expandable metal member **220**. The reactive fluid **510**, in the illustrated embodiment, causes the expandable metal member **220** to expand in response to hydrolysis, and thereby form an expanded metal member **530**. In accordance with the disclosure, given that the reactive fluid **510** now includes the colloid particles **440**, the expanded metal member **530** includes the colloid particles **440** in interstitial spaces thereof.

[0054] Turning to FIG. **6**, illustrated is a well system **600** designed, manufactured and/or operated according to one or more alternative embodiments of the disclosure. The well system **600** is similar in many respects to the well system **100** of FIG. **1**. Accordingly, like reference numbers have been used to indicate similar, if not identical, features. The well system **600** differs, for the most part, from the well system **100**, in that the well system **600** employs a different downhole tool **680**. For instance, wherein the downhole tool **180** itself includes the colloid particles surrounding a surface of the expandable metal member, the downhole tool **680** does not.

[0055] Turning to FIG. **7**, illustrated is one embodiment of a downhole tool **700** designed, manufactured and/or operated according to one or more embodiments of the disclosure, as might be used in a well system such as the well system **600** of FIG. **6**. The downhole tool **700** of FIG. **7**, in one or more embodiments, is similar to the downhole tool **200** of FIG. **2**. Accordingly, like reference numbers have been used to indicated similar, if not identical, features. The downhole tool **700** differs, for the most part, from the downhole tool **200**, in that the downhole tool **700** itself does not include colloid particles surrounding a surface of the expandable metal member. Furthermore, the dissolvable sleeve **260** of the embodiment of FIG. **2** is thus not necessary. As will be understood below, the colloid particles will be introduced near the expandable metal member **220** using fluid within the wellbore.

[0056] Turning now to FIGS. **8A** through **8F**, illustrated are various different views of a well system **800** and downhole tool **880** designed, manufactured and/or operated according to one or more embodiments of the disclosure at different stages of deployment. The well system **800** of FIGS. **8A** through **8F** is similar, in many respects, to the well system **700** of FIG. **7**. Furthermore, the downhole tool **880** of FIGS. **8A** through **8F** is similar, in many respects, to the downhole tool **700** of FIG. **7**. Accordingly, like reference numbers have been used to indicate similar, if not identical, features.

[0057] Turning initially to FIGS. **8A** and **8B**, the downhole tool **880** is located within the wellbore **120** in a run-in-hole state. Further to the embodiment of FIGS. **8A** and **8B**, reactive fluid has yet to be located within the wellbore **120** and surrounding the downhole tool **880**, and thus the expandable metal member **220** has yet to expand in response to hydrolysis. While the embodiment of FIGS. **8A** and **8B** illustrates that the wellbore **120** is free of any fluid, other embodiments may exist wherein the downhole tool **880** is surrounding by non-reactive fluid (e.g., oil based mud, etc.).

[0058] Turning now to FIGS. **8C** and **8D**, illustrated are the well system **800** and downhole tool **880** of FIGS. **8A** and **8B**, after a reactive colloidal dispersion of colloid particles **810** (e.g., as described above) is starting to fill the wellbore **120**, but has yet to reach the downhole tool **880**. As reactive fluid (e.g., the reactive colloidal dispersion of colloid particles **810**) has yet to reach the downhole tool **880**, the expandable metal member **220** has again yet to expand in response to hydrolysis.

[0059] Turning now to FIGS. **8E** and **8F**, illustrated are the well system **800** and downhole tool **880** of FIGS. **8C** and **8D**, after the reactive colloidal dispersion of colloid particles **810** has filled the wellbore at least past the downhole tool **880**. Accordingly, at this stage the downhole tool **880**, and more particularly the expandable metal member **220**, is in contact with the reactive colloidal dispersion of colloid particles **810**. The reactive fluid within the reactive colloidal dispersion of colloid particles **810** causes the expandable metal member **220** to expand in response to hydrolysis, and thereby form an expanded metal member **885**. In accordance with the disclosure, given that the reactive colloidal dispersion of colloid particles **810** includes the colloid particles, the expanded metal member **885** benefits from the inclusion of the colloid particles **890** in interstitial spaces

thereof. Accordingly, the expanded metal member **885** having the colloid particles **890** is more impervious to fluid and/or gas migration therethrough, as compared to a traditional expanded metal member.

[0060] Turning now to FIGS. **9A** through **9J**, illustrated are various different views of a well system **900** and downhole tool **980** designed, manufactured and/or operated according to one or more embodiments of the disclosure at different stages of deployment. The well system **900** of FIGS. **9A** through **9J** is similar, in many respects, to the well system **700** of FIG. **7**. Furthermore, the downhole tool **980** of FIGS. **9A** through **9J** is similar, in many respects, to the downhole tool **700** of FIG. **7**. Accordingly, like reference numbers have been used to indicate similar, if not identical, features.

[0061] Turning initially to FIGS. **9A** and **9B**, the downhole tool **980** is located within the wellbore **120** in a run-in-hole state. Further to the embodiment of FIGS. **9A** and **9B**, reactive fluid has yet to be located within the wellbore **120** and surrounding the downhole tool **980**, and thus the expandable metal member **220** has yet to expand in response to hydrolysis. While the embodiment of FIGS. **9A** and **9B** illustrates that the wellbore **120** is free of any fluid, other embodiments may exist wherein the downhole tool **980** is surrounded by non-reactive fluid (e.g., oil based mud, etc.).

[0062] Turning now to FIGS. **9C** and **9D**, illustrated are the well system **900** and downhole tool **980** of FIGS. **9A** and **9B**, after non-reactive fluid **905** is starting to fill the wellbore **120**, but has yet to reach the downhole tool **980**. As reactive fluid has yet to reach the downhole tool **980**, the expandable metal member **220** has again yet to expand in response to hydrolysis. Furthermore, the non-reactive fluid **905** may or may not include the colloid particles. Nevertheless, in the embodiment of FIGS. **9C** and **9D**, the non-reactive fluid **905** does not include the colloid particles.

[0063] Turning now to FIGS. **9E** and **9F**, illustrated are the well system **900** and downhole tool **980** of FIGS. **9C** and **9D**, after non-reactive fluid **905** continues to fill the wellbore **120** and has reached the downhole tool **980**. Further to the embodiment of FIGS. **9E** and **9F**, a slug of the reactive colloidal dispersion of colloid particles **910** has been included within the wellbore **120**. For example, in the embodiment shown, the slug of the reactive colloidal dispersion of colloid particles **910** is located between the downhole slug of non-reactive fluid **905**, and an uphole slug of fluid **915**. The uphole slug of fluid **915**, in the embodiment of FIGS. **9E** and **9F**, does not include the colloid particles, and may or may not be a reactive fluid. The uphole slug of fluid **915** also may or may not be a similar fluid as the non-reactive fluid **905**.

[0064] Turning now to FIGS. **9G** and **9H**, illustrated are the well system **900** and downhole tool **980** of FIGS. **9E** and **9F**, after the slug of the reactive colloidal dispersion of colloid particles **910** has just reached the downhole tool **980**, but has not had enough time to start the hydrolysis process. In at least one embodiment, the downhole tool **980** has a length (l), and the slug of the reactive colloidal dispersion of colloid particles **910** has a thickness (t) greater than the length (l) of the downhole tool **980**. In at least one other embodiment, the slug of the reactive colloidal dispersion of colloid particles **910** has a thickness (t) less than 10 times the length (l) of the downhole tool **980**. In yet one other embodiment, the slug of the reactive colloidal dispersion of colloid particles **910** has a thickness (t) less than 3 times the length (l) of the downhole tool **980**, if not less than 1.5 times the length (l) of the downhole tool **980**.

[0065] One benefit of using the slug of the reactive colloidal dispersion of colloid particles **910** is that a smaller volume of the colloidal dispersion of colloid particles is required than the embodiment of FIGS. **8A** through **8F** wherein the entire wellbore **120** down to the expandable metal member **220** is filled with the colloidal dispersion of colloid particles. Thus, doing such may be a great cost savings.

[0066] Turning now to FIGS. **9I** and **9J**, illustrated are the well system **900** and downhole tool **980** of FIGS. **9G** and **9H**, after the reactive fluid within the slug of the reactive colloidal dispersion of colloid particles **910** causes the expandable metal member **220** to expand in response to hydrolysis, and thereby form an expanded metal member **985**. In accordance with the disclosure, given that the

slug of the reactive colloidal dispersion of colloid particles **910** includes the colloid particles, the expanded metal member **985** benefits from the inclusion of the colloid particles **990** in interstitial spaces thereof. Accordingly, the expanded metal member **985** having the colloid particles **990** is more impervious to fluid and/or gas migration therethrough, as compared to a traditional expanded metal member.

[0067] Aspects disclosed herein include: [0068] A. A downhole tool, the downhole tool including: 1) a housing; 2) an expandable metal member positioned about the housing, the expandable metal member comprising a metal configured to expand in response to hydrolysis; and 3) colloid particles surrounding a surface of the expandable metal member. [0069] B. A well system, the well system including: 1) a wellbore positioned within a subterranean formation; 2) a downhole tool positioned within the wellbore, the downhole tool including: a) a housing; b) an expandable metal member positioned about the housing, the expandable metal member comprising a metal configured to expand in response to hydrolysis; and 3) colloid particles surrounding a surface of the expandable metal member. [0070] C. A method, the method including: 1) positioning a downhole tool within a wellbore of a subterranean formation, the downhole tool including: a) a housing; b) an expandable metal member positioned about the housing, the expandable metal member comprising a metal configured to expand in response to hydrolysis; and c) colloid particles surrounding a surface of the expandable metal member; 2) subjecting the expandable metal member to a reactive fluid while in the presence of the colloid particles, thereby forming an expanded metal member having the colloid particles in interstitial spaces thereof. [0071] D. A well system, the well system including: 1) a wellbore positioned within a subterranean formation; 2) a downhole tool positioned within the wellbore, the downhole tool including: a) a housing; and b) an expandable metal member positioned about the housing, the expandable metal member comprising a metal configured to expand in response to hydrolysis; and c) a reactive colloidal dispersion of colloid particles surrounding a surface of the expandable metal member. [0072] E. A method, the method including: 1) positioning a downhole tool within a wellbore of a subterranean formation, the downhole tool including: a) a housing; and b) an expandable metal member positioned about the housing, the expandable metal member comprising a metal configured to expand in response to hydrolysis; and 2) positioning a reactive colloidal dispersion of colloid particles about a surface of the expandable metal member, thereby forming an expanded metal member having the colloid particles in interstitial spaces thereof.

[0073] Aspects A, B, C, D and E may have one or more of the following additional elements in combination: Element 1: further including a dissolvable sleeve positioned about the colloid particles, the dissolvable sleeve configured to keep the colloid particles proximate the expandable metal member until such time as the expandable metal member is to undergo the hydrolysis. Element 2: wherein the colloid particles are a powder of colloid particles, the dissolvable sleeve holding the powder of colloid particles proximate the expandable metal member until such time as the expandable metal member is to undergo the hydrolysis. Element 3: wherein the colloid particles form a part of a non-reactive colloidal dispersion of colloid particles, the dissolvable sleeve holding the non-reactive colloidal dispersion of colloid particles proximate the expandable metal member until such time as the expandable metal member is to undergo the hydrolysis. Element 4: wherein the non-reactive colloidal dispersion of colloid particles is a sturdy non-reactive colloidal dispersion of colloid particles. Element 5: wherein the non-reactive colloidal dispersion of colloid particles is an extremely sturdy non-reactive colloidal dispersion of colloid particles. Element 6: wherein the non-reactive colloidal dispersion of colloid particles is an excessively sturdy non-reactive colloidal dispersion of colloid particles. Element 7: wherein the non-reactive colloidal dispersion of colloid particles is a combination of a non-reactive fluid and the colloid particles. Element 8: wherein the non-reactive fluid is oil. Element 9: wherein the colloid particles are small colloid particles. Element 10: wherein the colloid particles are extremely small colloid particles. Element 11: wherein the colloid particles are excessively small colloid particles. Element 12:

wherein the colloid particles are latex colloid particles. Element 13: wherein the colloid particles are clay colloid particles. Element 14: further including a dissolvable sleeve positioned about the colloid particles, the dissolvable sleeve configured to keep the colloid particles proximate the expandable metal member until such time as the expandable metal member is to undergo the hydrolysis, and further wherein the subjecting dissolves the dissolvable sleeve and allows the reactive fluid to encounter the expandable metal and form the expanded metal. Element 15: wherein the colloid particles are a powder of colloid particles, the dissolvable sleeve holding the powder of colloid particles proximate the expandable metal member until such time as the expandable metal member is to undergo the hydrolysis. Element 16: wherein the subjecting causes the powder of colloid particles and the reactive fluid to form a reactive colloidal dispersion of colloid particles. Element 17: wherein the colloid particles form a part of a non-reactive colloidal dispersion of colloid particles, the dissolvable sleeve holding the non-reactive colloidal dispersion of colloid particles proximate the expandable metal member until such time as the expandable metal member is to undergo the hydrolysis. Element 18: wherein the subjecting causes the colloid particles in the non-reactive colloidal dispersion of colloid particles and the reactive fluid to form a reactive colloidal dispersion of colloid particles. Element 19: wherein the reactive colloidal dispersion of colloid particles is a sturdy reactive colloidal dispersion of colloid particles. Element 20: wherein the reactive colloidal dispersion of colloid particles is an extremely sturdy reactive colloidal dispersion of colloid particles. Element 21: wherein the reactive colloidal dispersion of colloid particles is an excessively sturdy reactive colloidal dispersion of colloid particles. Element 22: wherein the colloid particles are ionically charged colloid particles. Element 23: wherein positioning the reactive colloidal dispersion of colloid particles about the surface of the expandable metal member includes filling the wellbore above the downhole tool with the reactive colloidal dispersion of colloid particles. Element 24: wherein positioning the reactive colloidal dispersion of colloid particles about the surface of the expandable metal member includes filling the wellbore with a slug of the reactive colloidal dispersion of colloid particles and a slug of fluid without the colloid particles. Element 25: wherein the slug of the reactive colloidal dispersion of colloid particles is pumped downhole prior to the slug of fluid without the colloid particles. Element 26: wherein the slug of fluid without the colloidal particles is an uphole slug of fluid without the colloidal particles, and further including pumping a downhole slug of fluid without the colloidal particles prior to pumping the slug of the reactive colloidal dispersion of the colloid particles. Element 27: wherein the downhole slug of fluid without the colloidal particles is a non-reactive slug of fluid. Element 28: wherein the downhole tool has a length (l), and further wherein the slug of the reactive colloidal dispersion of colloid particles has a thickness (t) greater than the length (l) of the downhole tool. Element 29: wherein the slug of the reactive colloidal dispersion of colloid particles has a thickness (t) less than 10 times the length (l) of the downhole tool. Element 30: wherein the slug of the reactive colloidal dispersion of colloid particles has a thickness (t) less than 3 times the length (l) of the downhole tool. Element 31: wherein the slug of the reactive colloidal dispersion of colloid particles has a thickness (t) less than 1.5 times the length (l) of the downhole tool.

[0074] Those skilled in the art to which this application relates will appreciate that other and further additions, deletions, substitutions and modifications may be made to the described embodiments.

Claims

1. A well system, comprising: a wellbore positioned within a subterranean formation; a downhole tool positioned within the wellbore, the downhole tool including: a housing; and an expandable metal member positioned about the housing, the expandable metal member comprising a metal configured to expand in response to hydrolysis; and a reactive colloidal dispersion of colloid particles surrounding a surface of the expandable metal member.

2. The well system as recited in claim 1, wherein the reactive colloidal dispersion of colloid particles is a sturdy reactive colloidal dispersion of colloid particles.
3. The well system as recited in claim 1, wherein the reactive colloidal dispersion of colloid particles is an extremely sturdy reactive colloidal dispersion of colloid particles.
4. The well system as recited in claim 1, wherein the reactive colloidal dispersion of colloid particles is an excessively sturdy reactive colloidal dispersion of colloid particles.
5. The well system as recited in claim 1, wherein the colloid particles are small colloid particles.
6. The well system as recited in claim 1, wherein the colloid particles are extremely small colloid particles.
7. The well system as recited in claim 1, wherein the colloid particles are excessively small colloid particles.
8. The well system as recited in claim 1, wherein the colloid particles are latex colloid particles.
9. The well system as recited in claim 1, wherein the colloid particles are clay colloid particles.
10. The well system as recited in claim 1, wherein the colloid particles are ionically charged colloid particles.
11. A method, comprising: positioning a downhole tool within a wellbore of a subterranean formation, the downhole tool including: a housing; and an expandable metal member positioned about the housing, the expandable metal member comprising a metal configured to expand in response to hydrolysis; and positioning a reactive colloidal dispersion of colloid particles about a surface of the expandable metal member, thereby forming an expanded metal member having the colloid particles in interstitial spaces thereof.
12. The method as recited in claim 11, wherein positioning the reactive colloidal dispersion of colloid particles about the surface of the expandable metal member includes filling the wellbore above the downhole tool with the reactive colloidal dispersion of colloid particles.
13. The method as recited in claim 11, wherein positioning the reactive colloidal dispersion of colloid particles about the surface of the expandable metal member includes filling the wellbore with a slug of the reactive colloidal dispersion of colloid particles and a slug of fluid without the colloid particles.
14. The method as recited in claim 13, wherein the slug of the reactive colloidal dispersion of colloid particles is pumped downhole prior to the slug of fluid without the colloid particles.
15. The method as recited in claim 14, wherein the slug of fluid without the colloidal particles is an uphole slug of fluid without the colloidal particles, and further including pumping a downhole slug of fluid without the colloidal particles prior to pumping the slug of the reactive colloidal dispersion of the colloid particles.
16. The method as recited in claim 15, wherein the downhole slug of fluid without the colloidal particles is a non-reactive slug of fluid.
17. The method as recited in claim 13, wherein the downhole tool has a length (l), and further wherein the slug of the reactive colloidal dispersion of colloid particles has a thickness (t) greater than the length (l) of the downhole tool.
18. The method as recited in claim 17, wherein the slug of the reactive colloidal dispersion of colloid particles has a thickness (t) less than 10 times the length (l) of the downhole tool.
19. The method as recited in claim 17, wherein the slug of the reactive colloidal dispersion of colloid particles has a thickness (t) less than 3 times the length (l) of the downhole tool.
20. The method as recited in claim 17, wherein the slug of the reactive colloidal dispersion of colloid particles has a thickness (t) less than 1.5 times the length (l) of the downhole tool.
21. The method as recited in claim 11, wherein the colloid particles are ionically charged colloid particles.
22. The method as recited in claim 11, wherein the reactive colloidal dispersion of colloid particles is a sturdy reactive colloidal dispersion of colloid particles.
23. The method as recited in claim 11, wherein the reactive colloidal dispersion of colloid particles

is an extremely sturdy reactive colloidal dispersion of colloid particles.

24. The method as recited in claim 11, wherein the reactive colloidal dispersion of colloid particles is an excessively sturdy reactive colloidal dispersion of colloid particles.

25. The method as recited in claim 11, wherein the colloid particles are small colloid particles.

26. The method as recited in claim 11, wherein the colloid particles are extremely small colloid particles.

27. The method as recited in claim 11, wherein the colloid particles are excessively small colloid particles.

28. The method as recited in claim 11, wherein the colloid particles are latex colloid particles.

29. The method as recited in claim 11, wherein the colloid particles are clay colloid particles.
